

Course Number	0418-211	Course Title	Theory of Computation I
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Ludovit Niepel
Compulsory	Yes	Prerequisite	0418-111, 0418-201

1

Textbook

<i>Title:</i>	<i>An Introduction to Formal Languages and Automata</i>
<i>Author:</i>	<i>Peter Linz</i>
<i>Publisher:</i>	<i>Jones and Bartlett Publishers, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Basic concepts: Alphabets, strings, languages; concatenation; powers and closure of strings and languages; phrase-structure grammars.	6
2	Regular languages: Regular grammars, regular expressions, deterministic and nondeterministic finite automata and their equivalence; minimization of deterministic finite automata; basic properties of regular languages.	10
3	Context-free languages: Context-free grammars, nondeterministic pushdown automata and their equivalence to context-free grammars; basic properties of context-free languages; deterministic context-free languages; elements of parsing.	9
4	Higher classes of the Chomsky hierarchy: Turing machines; context-sensitive languages and their linear-bounded automata; recursively enumerable languages; computability and decidability; computational complexity.	9
5	Fundamentals of intelligent systems and Knowledge Based Reasoning Optimal vs. human-like reasoning, the Turing test, resolution and theorem proving, Probabilistic reasoning, Bayes theorem.	5
6	Exams.	3
	Total.	42

Grading

20%	Midterm 1
20%	Midterm 2

20%	Hwk/quiz/project
40%	Final

Course outcomes

Students should be able to:

1. Design and explain regular languages in different notations: DFA, NFA, regular expressions and grammars, and convert between these notations.
2. Use the pumping lemma to identify not regular languages.
3. Understand pushdown automata, context-free grammars, and their normal forms, and apply a membership algorithm for context-free languages.
4. Understand the classification of languages based on the Chomsky hierarchy and the respective properties of language classes.
5. Understand the Turing thesis and its implications, and explain basic concepts of the theory of decidability.
6. Explain the operation of the resolution technique for theorem proving and role of Bayesian reasoning in computer science.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1.	L					
2.	L					
3.	M					
4.	H					
5.	M		M			
6.	L					